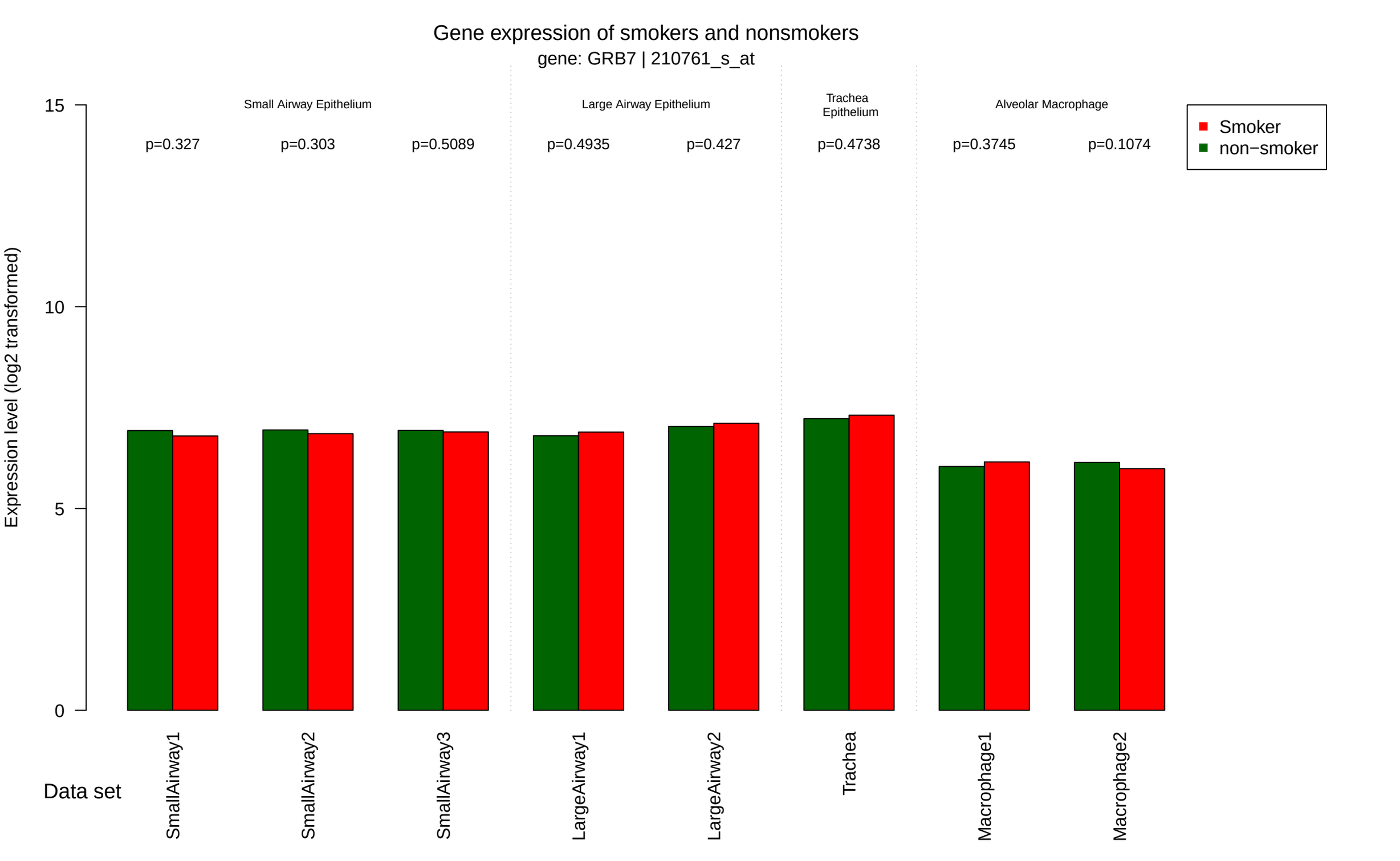
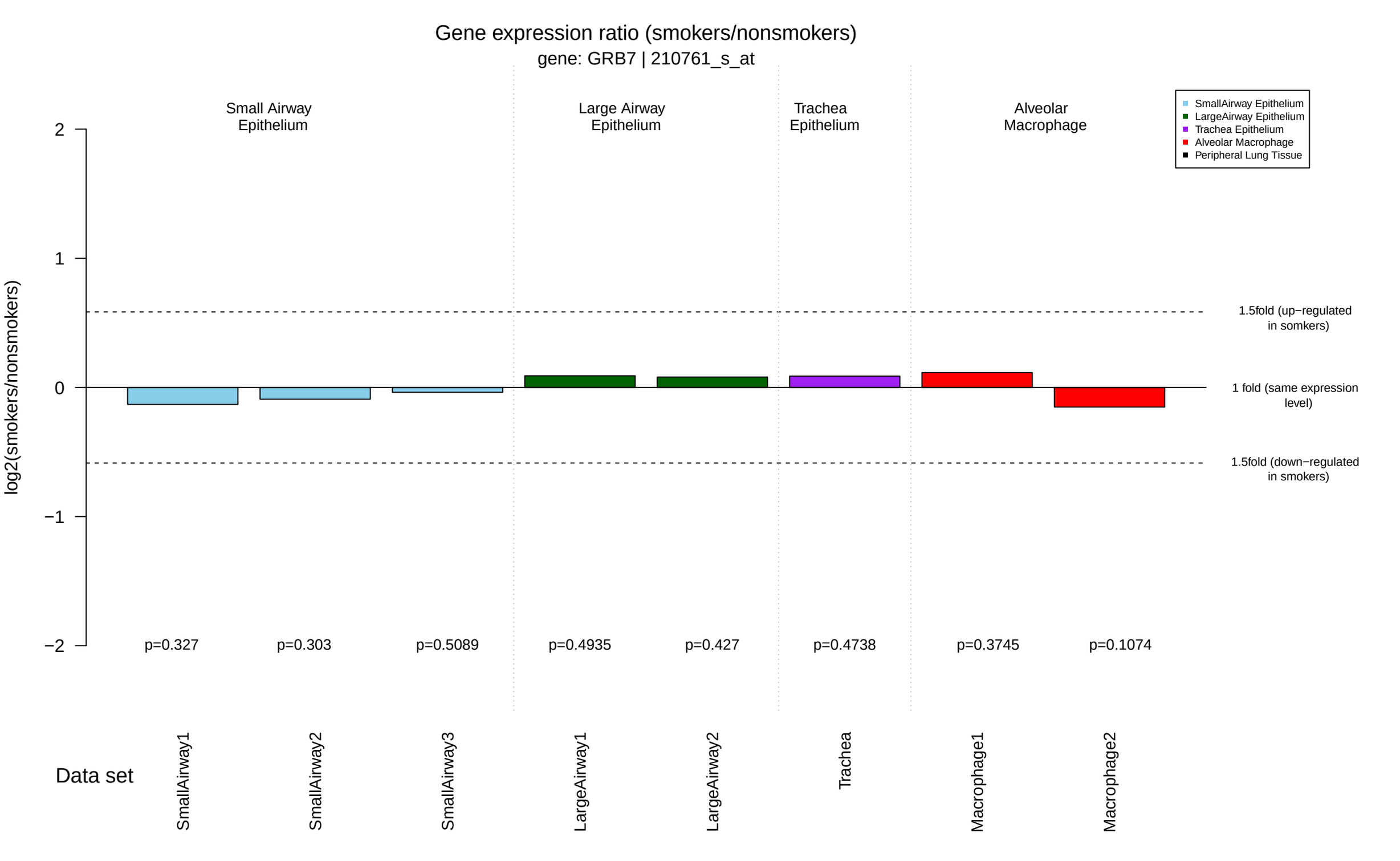
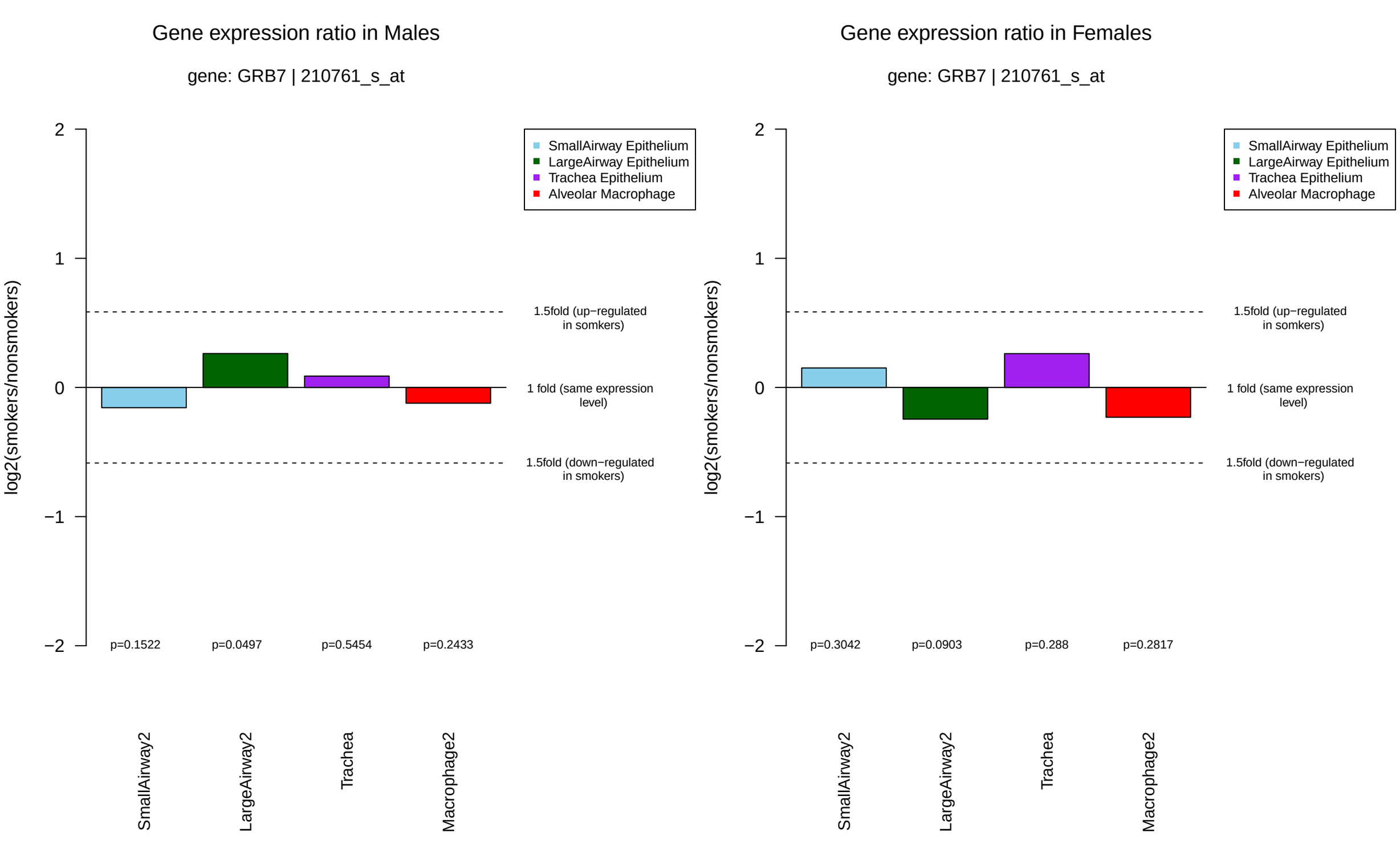
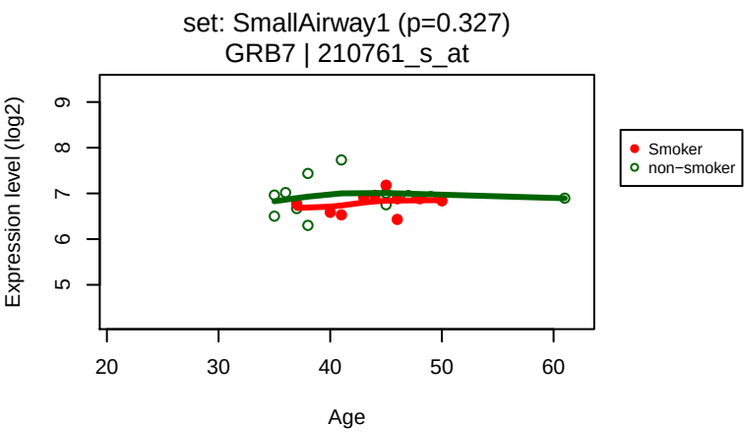
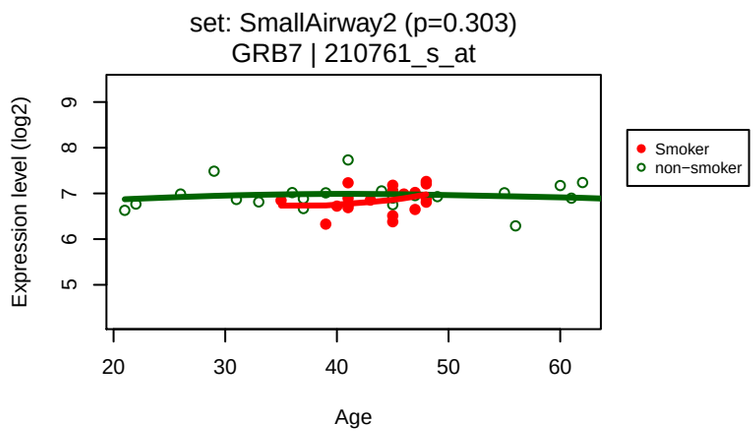
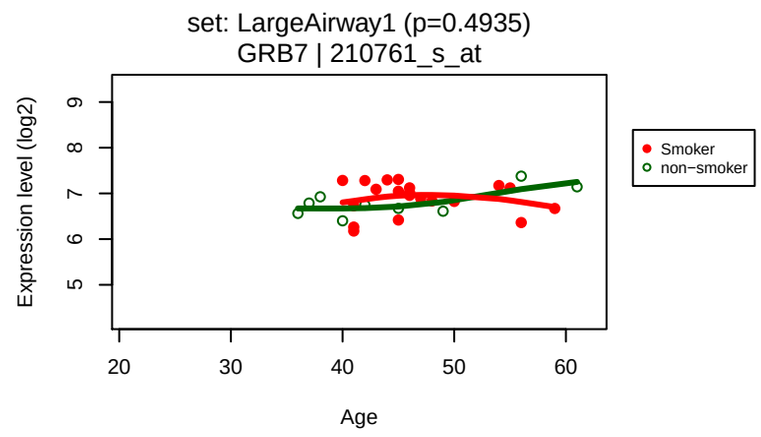

Gene expression of smokers and nonsmokers
gene: GRB7 | 210761_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.327
p=0.303
p=0.5089
p=0.4935
p=0.427
p=0.4738
p=0.3745
p=0.1074
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

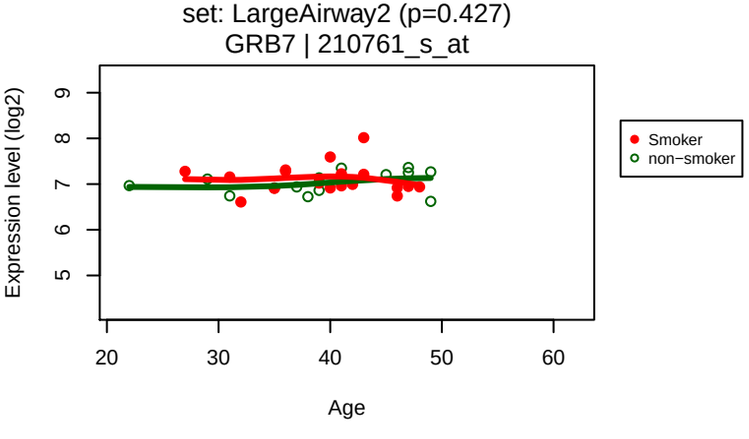

Gene expression ratio (smokers/nonsmokers)
gene: GRB7 | 210761_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.327
p=0.303
p=0.5089
p=0.4935
p=0.427
p=0.4738
p=0.3745
p=0.1074
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

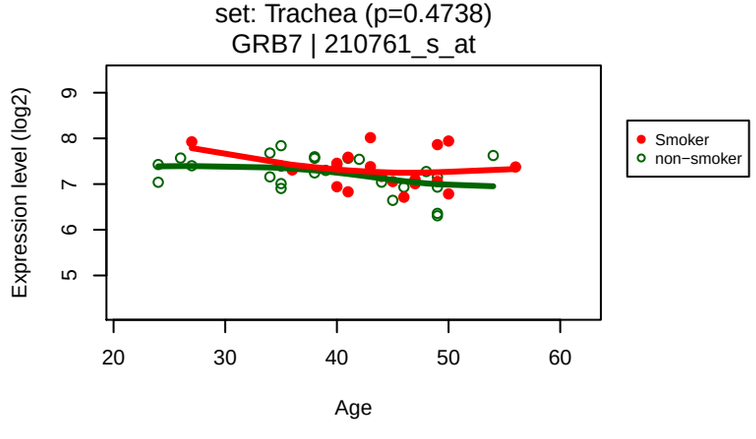

Gene expression ratio in Males
gene: GRB7 | 210761_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1522
p=0.0497
p=0.5454
p=0.2433
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRB7 | 210761_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3042
p=0.0903
p=0.288
p=0.2817
SmallAirway2
LargeAirway2
Trachea
Macrophage2

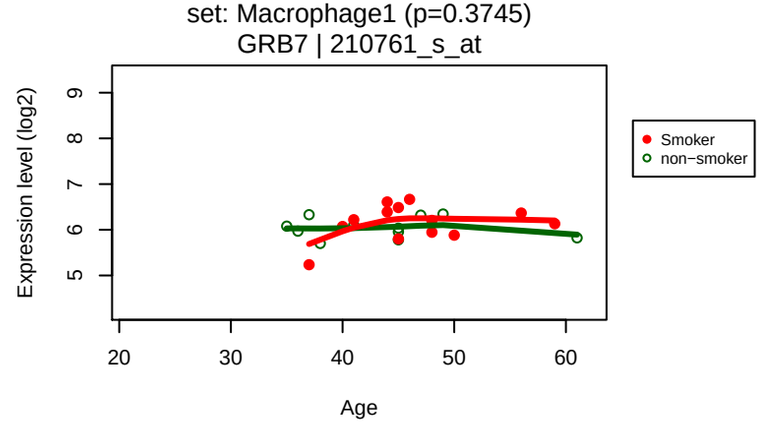

set: SmallAirway1 (p=0.327)
GRB7 | 210761_s_at
Expression level (log2)
Age
Smoker
non−smoker

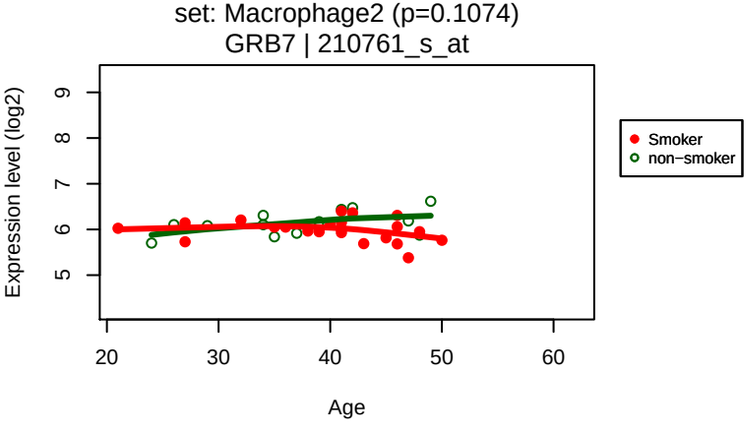

set: SmallAirway2 (p=0.303)
GRB7 | 210761_s_at
Expression level (log2)
Age
Smoker
non−smoker

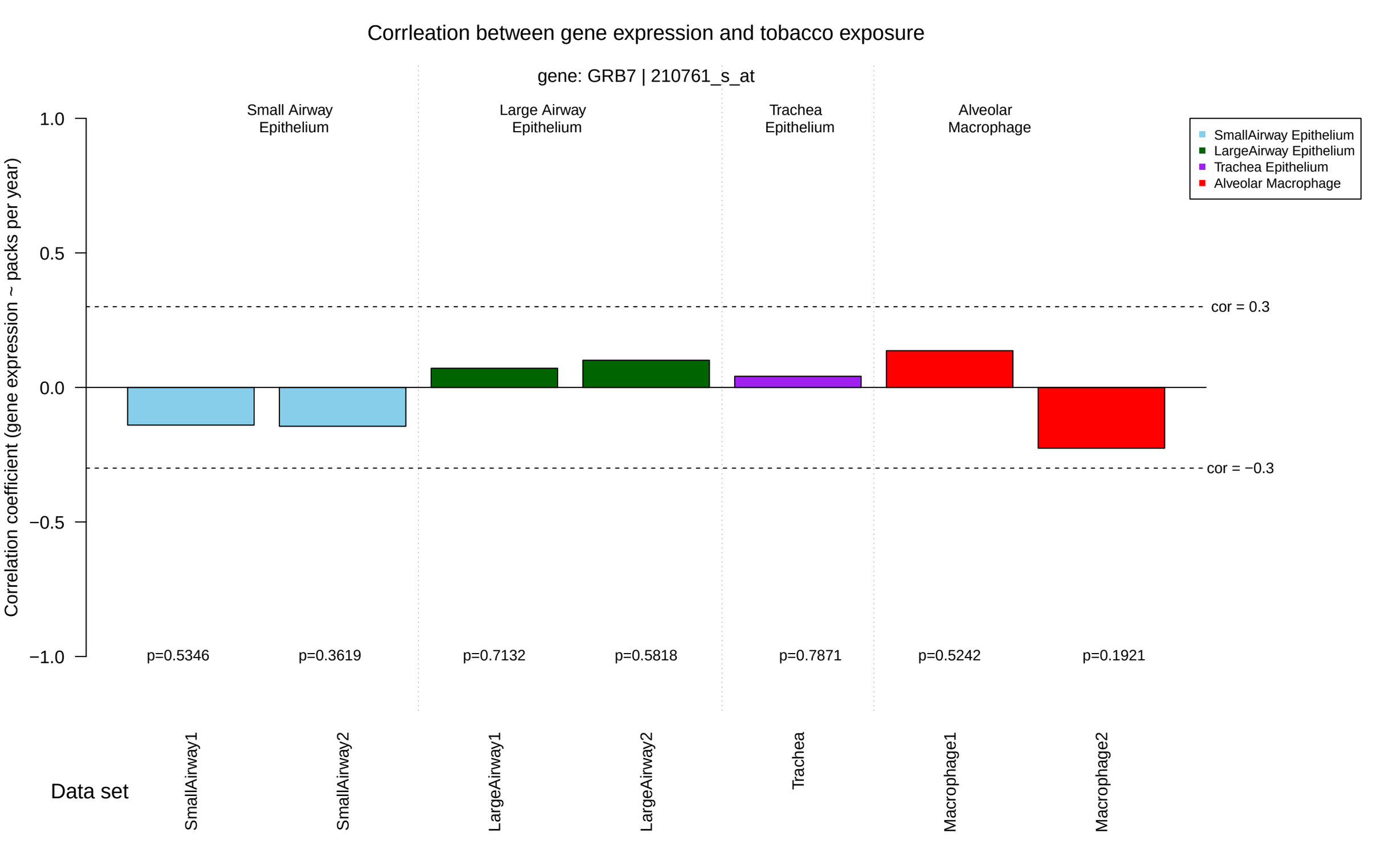

set: LargeAirway1 (p=0.4935)
GRB7 | 210761_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.427)
GRB7 | 210761_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4738)
GRB7 | 210761_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3745)
GRB7 | 210761_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1074)
GRB7 | 210761_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRB7 | 210761_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5346
p=0.3619
p=0.7132
p=0.5818
p=0.7871
p=0.5242
p=0.1921
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2